

Scientific Journals and Conferences

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Research is a Social Activity

A key feature of research is **social interaction**.

- ① New ideas need to be critically discussed with other researchers
- ② New knowledge needs to be "accepted" and "adopted" by other researchers

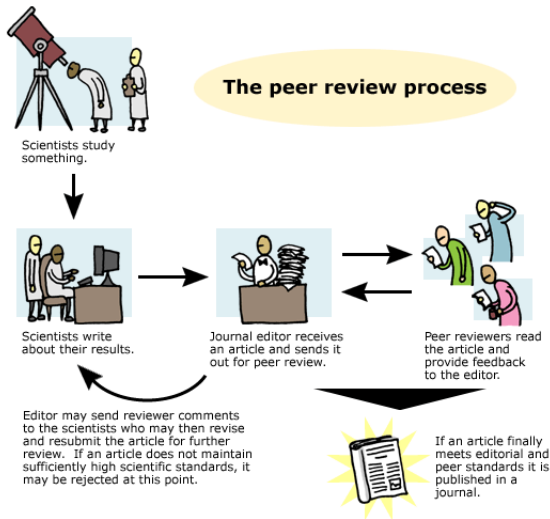
Research is **communicated** through **scientific articles**:

- **Conference** (mainly for discussion)
- **Journals** (mainly for acceptance and adoption).

Scientific Articles (a.k.a. **papers**)

- Intended to further the **progress** of science
- Reporting **new** research
- **Written** by scientists instead of professional journalists.
- **Read** by scientists in the field
- (peer) **reviewed** by scientists
- Ruled with high **ethical** standards (highly based on **trust**)

Peer Reviewing



Peer reviewing

Peer reviewing is the standard method to filter out bad research.

- When a researcher submits a paper to a journal several **colleagues** (called **reviewers** or **referees**) assess its quality
 - **Non-blind** (all names are known)
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- Peer Reviewing is
 - Time Consuming
 - Difficult
 - Non-rewarded (it is done **out of duty**)
- The next lecture we will cover it

Does Peer Reviewing work?

Bohannon's experiment (2013)

60% of fake papers were accepted in (fee-charging) journals

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Why does not it work better?

- Too much (unrewarded, uncredited) work.
- Lack of professionalism from reviewers.
- Authors incentives are missaligned with Scientific Attitude
- It is difficult to assess what is worth to publish.

An Inflationary Model?

There are way too many papers published every year

Source: Freakonomics

4 million articles are published in somewhere between 25 and 50000 journals

There has been an exponential growth in the last decades

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- Publishers want researchers to publish (Global scholarly publishing is a \$28000 million market)
- Governments want their scientists to publish (accountability)
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It is important to know what to read and where to publish

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- How should we sit on the toilet?



Should we take the phone to the toilet?

PublishAboutBrowse

 OPEN ACCESS  PEER-REVIEWED

RESEARCH ARTICLE

Smartphone use on the toilet and the risk of hemorrhoids

Chethan Ramprasad, Colin Wu , Jocelyn Chang , Vikram Rangan, Johanna Iturrino, Sarah Bailou, Prashant Singh, Anthony Lembo, Judy Nee, Trisha Pasricha 

Published: September 3, 2025 • <https://doi.org/10.1371/journal.pone.0329983>

Article	Authors	Metrics	Comments	Media Coverage
				

Abstract

Introduction

Methods

Results

Abstract

Smartphones are ubiquitous in daily life, with many people now using them while sitting on the toilet. Despite anecdotal evidence that length of time spent on the toilet is a risk factor for hemorrhoids, a multivariate analysis of smartphone use has not been performed. This study

- A conference is a physical event in which researchers present orally their work to colleagues
- The presentation is based on a previously submitted (and accepted) paper
- All papers are published in the conference **proceedings**
- In most fields the acceptance criterion of conferences is soft
- In Informatics, some conferences have a very hard acceptance criterion
 - because it has been a very dynamic field
 - conferences provide a quick way to have results disseminated

Conferences and Journals serve different goals

Acceptance is based on a **peer review** process

- **Workshops**

- preliminary work (some times just prospective ideas)
- light review process
- the emphasis is on discussion (having feedback from others)
- non-archival proceedings

- **Conferences**

- An atomic piece of original work (months of work)
- medium review process
- the emphasis is on discussion (convincing the other to take home the learned lesson)
- archival proceedings (**original** contribution)

- **Journals**

- A well presented substantial piece of original work (1 year of work)
- heavy review process
- the emphasis is on increasing knowledge corpus
- original work (may be extended from a conference)

- **Monographs** (comprehensive presentation of several years of coherent work by one or a few groups)
- **Handbooks** (compilation of main results on main sub-topics) → help newcomers to the field
- **Textbooks** (it somehow proves that the topic is consolidated) → can be taught at graduate and master level

Peer Review: conferences vs journals

Journals:

- 3-5 reviewers
- unbounded number of reviewing iterations (typically 1 or 2)
- Slow turnaround (from months to years), although its getting faster.

Conferences:

- 2-3 reviewers
- 0 or 1 reviewing iterations
- Time constraints
- Fast turnaround (1-2 months)

Types of Journals: Scope

- The **scope** of a Journal is the set of topics that it covers
- There are journals with wider and narrower scopes
 - Very wide (*Science*), wide (*J. of the ACM*), narrow (*J. of Educational Technology in Higher Education*)
 - Papers published on **wide scope** journals should report results of interest to a larger audience.
 - Doing Research of interest to a large community is nice, but...
 - ...it **does not** mean that research is better.

Types of Conferences: Scope

The same thing happens with Conferences

- General
 - Hundreds of presentations, thousands of attendees
 - Parallel sessions, satellite events
 - Many tracks
 - Best place to see (and even talk to) the most important researchers
- Specific
 - < 100 presentations, few hundred attendees
 - Best place to socialize (sometimes even with the most important researchers)

Conferences: types of submissions

Since Conferences are social gatherings

Conference organizers look for original ways to attract attendees and promote discussion and social interaction.

- full paper (10-15 pages)
- short paper (5 pages)
- (extended) abstract (1-2 pages)
- Specific tracks (e.g. applications, doctoral,...)

Attendance is mandatory (because high absenteeism ruins the whole idea and causes financial risk to the organizers)

How can we distinguish good from not so good conferences?

Rankings:

- <https://www.core.edu.au/conference-portal>
- <https://gii-grin-scie-rating.scie.es/>

Other criteria

- Where respectful people publishes regularly
- Reasonable acceptance rate (below 60 percent)

Beware of fake conferences

Fake conference

A (very successful) business model where the researcher pays for:

- an accepted paper
- a short vacation

while everything is made to **pretend** that Science matters

Fact

Journals have two parallel structures: Scientific and Editorial

Journal (scientific) structure:

- *Editor-in-Chief*: Ultimate responsibility
- *Area editors*: Responsible for all the editorial process of assigned papers
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All these people are researchers affiliated to research institutions and do this work without any economic compensation.

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 - Typically do not have a private editorial behind
 - Freely available on-line
 - There are several sponsorship models (institutions, authors,...)
 - **Critics:** may lack professionalism (there is no clear evidence)
 - **Note: Repositories** are a different thing (pre-prints without peer review)

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- Many editorials are also adopting this model asking authors to pay a fee for publication.

Beware Fake (predatory) Journals

Predatory Journals

charging publication fees (\$500-\$2000) to authors without providing the editorial and publishing services associated with legitimate journals

- Exponential growth (from 50000 to 400000 papers in 4 years)
- It is not easy to characterize and report them.

[More details here](#)

Fact

There are good (reliable) and bad (unreliable) journals. The frontier is fuzzy and well populated.

Some guidelines to identify good journals:

- Scientific staff:
 - rotates periodically
 - affiliated to prestigious institutions
 - (truly and honestly) international
- Authors: prestigious researchers publish good work

Journal Quality (Quantitative elements))

Fact

Quantifying journal quality is impossible, but metrics are useful.

- **Citations:** It is generally believed that good papers are highly cited
 - Critiques: negative references, hot topics, self-references, group-of-friends references...
- **Impact factor (IF):**
 - average number of citations received per paper published in that journal during the two preceding years
 - used to compare the **quality** different **journals within** a certain field
 - for those journals that are indexed in the *Journal Citation Reports* (JCR) (Thomson Reuters)
- **Journal Citation Ranking (Quartile):** Journals of each field are ordered by IF. First quartile journals (Q1) are the 25% ranked first.

Evaluations

Researchers, institutions,..., are usually evaluated by their **number** of published papers taking into account the **number of citations** and the **IF** of the journals where the papers are published

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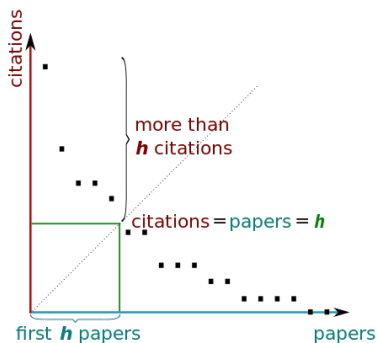
The use of IF has many critiques:

- based on arithmetic mean,
- unstable along time
- editors play the game
- opaque (JCR controlled by Thomsom)
- moderate correlation with expert's opinion
- Useless when comparing different disciplines
- In CS, people often references conference papers

Improved metrics

h-index

Metric proposed by Jorge E. Hirsch (2005, physicist at UCSD) that has become a usual complement to other metrics



- Most popular author-level metric
- Measures number of impact papers

On the use of Quantitative Indicators

Quantitative Indicators are useful, but problematic

Goodhart's Law

When a measure becomes a target, it ceases to be a good measure

D.T. Campbell 76

The more any quantitative social indicator is used for social decision-making, the more subject it will be to corruption pressures and the more apt it will be to distort and corrupt the social processes it is intended to monitor

Example from University Rankings:

- Northeastern University rose from #162 to #42 in US News ranking by aesthetic changes
- several US Colleges have been caught cheating

On the use of Quantitative Indicators

Quantitative Indicators have been (and are being) overused.

T. McGlynn (at realscientists, 2015)

I have been in a number of committees. I do not remember anybody looking at anybody's papers. Number and IF of publications are what counts

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- Do not base the evaluation on **quantitative indicators**
 - such as journal-based metrics like Impact Factor
 - *h*-index
- Consider **qualitative indicators** of value and impact
 - **narrative Curriculum Vita** (CV is narrative of impact instead of an enumeration of items)

A Wrapping up Story

Listen to this podcast (mandatory):

<https://open.spotify.com/episode/3HHQhvXjE8XmYCob3NjYf5>

And you will see

- The interesting life of **Linus Pauling**. Twice Nobel Prize and often said to be one of the greatest scientists of the 20th century.
- The **value of Research**
- The value and usefulness of **Ethics** (and the cost of morality)
- The **irrationality** of humans (even geniuses can be stubbornly biased)
- the robustness of the **Scientific Method**
- the **difficulty of knowledge discovery**
- And finally, **that you should not bother to drink orange juice for health**